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PHY 312 — Past Questions & Materials
by Renance

Course: PHY 312
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 6
Levels: 300

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Paper 1 of 6 — PHY 312 Electricity and Magnetism II

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF PHYSICS
FIRST SEMESTER EXAMINATION.

Course Code: PHY 312
Course Title: ELECTRICITY AND MAGNETISM II
Course Unit: 3 UNITS
Session: 2022/2023
Time Allowed: 2hrs 30 min
Instruction: Answer question 1, One in Section A and Two in Section B

Date: 20/05/2023

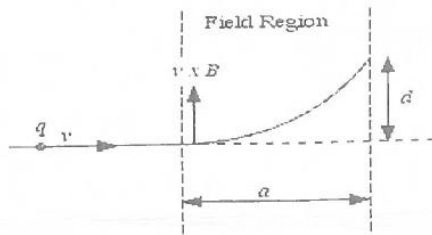
1. Explain Lorentz force law
2. Write an equation of Lorentz force law and define all the parameters
3. State (i) Maxwell equations for magnetostatics
(ii) Maxwell equations for electrostatics
and hence explain the difference between magnetic field and electric field.
(b) State the three fundamental quantities of magnetostatics
4. State the Biot- Savart law for magnetic field
5. Explain Ferromagnetic material and give four (4) examples
6. State four configurations that can be handled by Ampere's law
7. With the magnetic field produced by a Steady current. Use Biot- Savart Law to explain.
8. Write the two equations for Surface and Volume currents that the Biot- Savart law can be written as.

SECTION A

2. (a) A phonograph record of radius R , carrying a uniform surface charge σ , is rotating at constant angular velocity ω . Find its magnetic dipole moment.
(b) State the difference between the Boundary condition for electric field and that of magnetic field
3. A circular loop of wire, with radius R , lies in the xy plane, centred at the origin, and carries a current I running counter clockwise as viewed from the positive z axis.
(i) What is its magnetic dipole moment?
(ii) What is its (approximate) magnetic field at points far from the origin?
(iii) Determine the exact solution for $\vec{B}$ on the positive z axis.

SECTION B

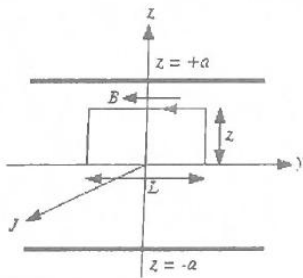
4. A particle of charge q enters the region of uniform magnetic field $\vec{B}$ (pointing into the page). The field deflects the particle a distance d above the original line of flight, as shown in diagram below.
(i) Is the charge positive or negative?
(ii) In terms of a , d , B , and q , find the momentum of the particle.



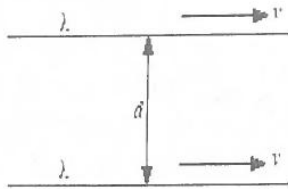
$$f = qvB$$

$$f = \frac{mv^2}{r}$$

5. A thick slab extending from $z = -a$ to $z = a$ carries a uniform volume current $\vec{J} = J\hat{i}$.
 (i) Find the magnetic field both inside and outside the slab.



6.



Suppose you have two infinite straight-line charges λ , a distance d apart, moving along at a constant v as above.

- (i) How fast would v have to be in order for the magnetic attraction to balance the electrical repulsion?

Paper 2 of 6 — PHY 312 - 2017/2018 - First Semester - ELECTRICITY AND MAGNETISM II

Level 300 • First Semester • PastQuestion

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COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS

1st SEMESTER EXAMINATION (2017/2018 SESSION)

PHY 312 (ELECTRICITY AND MAGNETISM I) 3Unit)

Instruction: Answer only question in Section A, One (1) question in Section B and. two (2) questions in section C Time: 2hrs 30 mins

SECTION A (25 marks)

1. (a) State (i) Maxwell equations for magnetostatics

i (ii) Maxwell equations for electrostatics

and hence explain the difference between magnetic field and electric field.

(b) State the three fundamental quantities of magnetostatics

(c) State the difference between the Boundary condition for electric field and that of magnetic field

(d) In the diagram below, if $\mathbf{F}$ is irrotational, show that $(\nabla \times \mathbf{B})^2 = 0$, where $\mathbf{F}$ is the vector from the origin to the point in question, That is check that $\nabla \times \mathbf{F} = 0$ and $\nabla \cdot \mathbf{F} = F$.

(e) Which of these Magnetic fields can exist? Determine the current density that created the valid fields. What Vector Potential corresponds to these fields? ,

(i) $\mathbf{B}(\mathbf{r}) = \mathbf{e} \times \mathbf{r}$

(ii) $\mathbf{B}(\mathbf{r}) = \sin(kr)\mathbf{r}$ (r, θ, ϕ coordinate system)

(f) Show that the magnetic field can not do any work on a particle, i.e. can not change its energy.

SECTION B (15 Marks each)

2. A circular loop of wire, with radius, X , lies in the xy plane, centred at the origin, and carries a current running counter clockwise as viewed from the positive z axis.

(i) What is its magnetic dipole moment?

(ii) What is its (approximate) magnetic field at points far from the origin?

Gii) Determine the exact solution for $\mathbf{B}$ on the positive z axis.

3. (a) Explain Lorentz covariance and hence otherwise state two distinct, but closely related meanings

(b) A phonograph record of radius R , carrying a uniform surface charge σ , is rotating at constant angular velocity ω .

(i) Find its magnetic dipole moment.

SECTION C (15 Marks each)

C4. A particle of charge q enters the region of uniform magnetic field $\mathbf{B}$ (pointing into the page). The field deflects the particle a distance d above the original line of flight, as shown in diagram below.

(i) Is the charge positive or negative? «

(ii) In terms of a , d , B , and q , find the momentum of the particle.

| Field Region |

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5. (a) With the magnetic field produced by a Steady current. Use Biot- Savart Law to explain.

(b) Write the two equations for Surface and Volume currents that the Biot- Savart law can be written as. i

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Suppose you have two infinite straight-line charges A , a distance d apart, moving along at a constant v as above.

(i) How fast would v have to be in order for the magnetic attraction to balance the electrical repulsion?

6. (a) A thick slab extending from $z = -a$ to $z = a$ carries a uniform volume current $\mathbf{J} = J\hat{i}$.

(i) Find the magnetic field both inside and outside the slab.

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(b) State four configurations that can be handled by Ampere's law. *

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Paper 3 of 6 — PHY 312 - 2016/2017 - First Semester - Electricity and Magnetism

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF PHYSICS
FIRST SEMESTER EXAMINATION

Academic Session: 2016/2017
Course Code: PHY 312
Course Title: Electricity and Magnetism II
Course Unit: 3 Unit
Time allowed: 2 hrs 30mins

Instruction: Answer any two (2) questions in section A and any two (2) in section B

SECTION A

QUESTION ONE

1 (a) A very long, uniform lines of charge with positive linear charge density $+\lambda$ lies along the x -axis. An identical line of charge lies along the y -axis.



(a) Determine the electric field $E(x,y)$ for all points in the x - y plane.
(b) Determine the change in electrostatic potential ΔV between the points $x=a, y=a$ and $x=a, y=3a$.
(c) Determine ΔV between the points $x=a, y=a$ and $x=3a, y=a$.
(d) How much work must be done to move a small negative charge $-q$ from the point $x=3a, y=3a$ to the point $x=a, y=a$?

(b) Which of these Magnetic fields can exist? Determine the current density that created the valid fields. What Vector Potential corresponds to these fields?

(i) $B(r) = e^{-r^2} \hat{x}$
(ii) $B(r) = e^{-r^2} \hat{y}$
(iii) $B(r) = \sin(kr) \hat{r}$ (r, θ, ϕ coordinate system)

QUESTION TWO

(a) Define Lorentz Covariance and explain the two distinct but closely related meanings.
(b) Maxwell's equation represent one of the most elegant and concise ways to state the fundamental of electricity and magnetism. Discuss.
(c) A circular loop of wire, with radius R , lies in the xy plane, centred at the origin, and carries a current I running counter clockwise as viewed from the positive z axis.
(i) What is its magnetic dipole moment?
(ii) What is its (approximate) magnetic field at points far from the origin?
(iii) Determine the exact solution for $\vec{B}$ on the positive z axis.

QUESTION THREE

(a) Show that the magnetic field can not do any work on a particle, i.e. can not change its energy.
(b) A spherical shell of conducting material has an inner radius a and outer radius b . The shell carries no net charge. Inside the shell, located at the centre, is a negative point charge of $-Q$.

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PHY 312 2016/2017 Page 1

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Q. The point charge is suspended from a thin thread and does not touch the shell. The coordinate r measures the distance from the centre of the sphere.



- Determine the charges on the inner and outer surfaces of the spherical shell.
- Determine the electric field everywhere in space. Sketch $E(r)$ vs. r .
- Determine the electrostatic potential $V(r)$ everywhere in space. Sketch $V(r)$ vs. r .
- Show that the relation $E_r = -dV/dr$ is satisfied.

SECTION B

QUESTION ONE

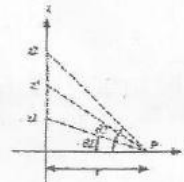
- Faradays' law is a fundamental relationship which comes from Maxwell's equations. Explain.
- Three very long parallel wires arranged as shown below. Two wires, which carry current I out of the page, pass through the points $y = -a$, $x = \pm a$. The third wire carries a current I into the page and is located at the point $x = 2a$, $y = -a$.



- Determine the magnetic field at the location of the top left wire ($x = 0$, $y = -a$) due to the other two wires.
- Determine the force per unit length on the top left wire.
- Determine the magnetic field $B_2(x)$ due to the two wires located at $x = 0$, $y = -a$ at an arbitrary point P on the axis, a distance x from the origin.
- Determine the magnetic field $B_3(x)$ at the point P due to the single wire at $x = 2a$, $y = -a$.
- From (c) and (d), determine the total magnetic field at the point P .

QUESTION TWO

- State and explain the three fundamental quantities of Magnetostatics.
- In the diagram below, if $\vec{B}$ is uniform, show that $\vec{A} = -(\vec{r} \times \vec{B})/2$, where $\vec{r}$ is the vector from the origin to the point in question. That is check that $\vec{\nabla} \times \vec{A} = \vec{B}$ and $\vec{\nabla} \cdot \vec{A} = 0$.



PHY 312 2016/2017
Page 2

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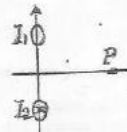
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- (c) The three quantities are related and if one of them is known, the other two can be calculated. Draw the table that summarises the relations between them.
 (d) State the difference between the Boundary condition for electric field and that of magnetic field.

QUESTION THREE

- (a) Explain Ferromagnetic material and give 4 (four) examples.
 (b) Two long, straight, parallel current-carrying wires are shown below. The wires are parallel to the x-axis (the +z-axis points out of the page). The current in the wire above the origin is flowing in the +z-direction ($I_1 = 3.2\text{A}$, out of the page), while the current in the wire below the origin is flowing in the -z-direction ($I_2 = 3.2\text{A}$, into the page). The wire above the origin passes through $y = +13.3\text{cm}$, the wire below the origin passes through $y = -13.3\text{cm}$, and the point P is at $x = +35.0\text{cm}$.



- (a) Find all three components of the magnetic field vector at the point P on the x-axis.
 (b) If an electron were travelling in the +z-direction at 5200m/s , what would be the magnitude and direction of the magnetic force acting on it as it passed through P?
 (c) What electric field (state all three components) would we need to apply to the electron in part (b) so that the net force acting on it were zero?

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PHY 312 2016/2017 Page 3

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Paper 4 of 6 — PHY 312 2017 2018

Level 300 • First Semester • PastQuestion

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COLLEGE OF SCIENCE
DEPARTMENT OF PHYSICS

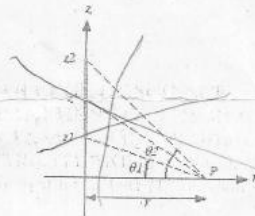
1ST SEMESTER EXAMINATION (2017/2018 SESSION)
PHY 312 (ELECTRICITY AND MAGNETISM II) (3Unit)

Instruction: Answer only question in Section A, One (1) question in Section B and two (2) questions in section C

Time: 2hrs 30 mins

SECTION A (25 marks)

1. (a) State (i) Maxwell equations for magnetostatics
(ii) Maxwell equations for electrostatics
and hence explain the difference between magnetic field and electric field.
(b) State the three fundamental quantities of magnetostatics
(c) State the difference between the Boundary condition for electric field and that of magnetic field
(d) In the diagram below, if $\vec{B}$ is uniform, show that $\vec{A} = -(\vec{r} \times \vec{B})/2$, where $\vec{r}$ is the vector from the origin to the point in question. That is check that $\nabla \times \vec{A} = \vec{B}$ and $\nabla \cdot \vec{A} = 0$.



- (e) Which of these Magnetic fields can exist? Determine the current density that created the valid fields. What Vector Potential corresponds to these fields?
(i) $\vec{B}(\vec{r}) = e^{-y^2} \hat{x}$
(iii) $B(r) = \sin(kr)r$ (r, θ, z coordinate system)
(f) Show that the magnetic field can not do any work on a particle, i.e. can not change its energy.

SECTION B (15 Marks each)

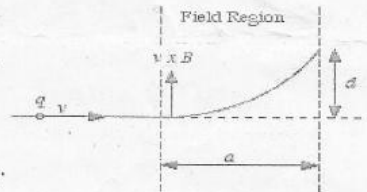
2. A circular loop of wire, with radius R , lies in the xy plane, centred at the origin, and carries a current I running counter clockwise as viewed from the positive z axis.
(i) What is its magnetic dipole moment?
(ii) What is its (approximate) magnetic field at points far from the origin?
(iii) Determine the exact solution for $\vec{B}$ on the positive z axis.
3. (a) Explain Lorentz covariance and hence otherwise state two distinct, but closely related meanings
(b) A phonograph record of radius R , carrying a uniform surface charge σ , is rotating at constant angular velocity ω .
(i) Find its magnetic dipole moment.

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SECTION C (15 Marks each)

4. A particle of charge q enters the region of uniform magnetic field $\vec{B}$ (pointing into the page). The field deflects the particle a distance d above the original line of flight, as shown in diagram below.
- Is the charge positive or negative?
 - In terms of a , d , B , and q , find the momentum of the particle.

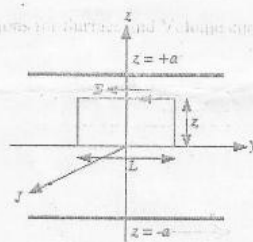


- With the magnetic field produced by a Steady current. Use Biot- Savart Law to explain.
 - Write the two equations for Surface and Volume currents that the Biot- Savart law can be written as.



Suppose you have two infinite straight-line charges λ , a distance d apart, moving along at a constant v as above.

- How fast would v have to be in order for the magnetic attraction to balance the electrical repulsion?
- A thick slab extending from $z = -a$ to $z = a$ carries a uniform volume current $\vec{J} = J\hat{i}$.
 - Find the magnetic field both inside and outside the slab.



- State four configurations that can be handled by Ampere's law.

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Paper 5 of 6 — Phy 312 Exam PQ ELECTRICITY AND MAGNETISM

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
SCHOOL OF POSTGRADUATE STUDIES
DEPARTMENT OF PHYSICS
FIRST SEMESTER EXAMINATION.

Course Code: PHY 312
Course Title: ELECTRICITY AND MAGNETISM II
Course Unit: 3 UNITS
Session: 2019/2020
Time Allowed: 2hrs 30 min

Date:

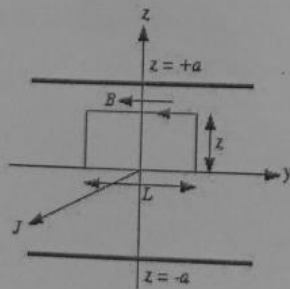
Instruction: Answer any two (2) questions in section A and any two (2) in section B

SECTION A

1. (a) Show that magnetic field can not do any work on a particle i.e. can not change its energy.
(b) Explain Lorentz force.
(b) A phonograph record of radius R , carrying a uniform surface charge σ , is rotating at constant angular velocity ω .
(i) Find its magnetic dipole moment.
2. (a) State (i) Maxwell equations for magnetostatics
(ii) Maxwell equations for electrostatics
and hence explain the difference between magnetic field and electric field.
(b) State the three fundamental quantities of magnetostatics
(c) State the difference between the Boundary condition for electric field and that of magnetic field
3. (a) State the Biot- Savart law for magnetic field
(b) A circular loop of wire, with radius R , lies in the xy plane, centred at the origin, and carries a current I running counter clockwise as viewed from the positive z axis.
(i) What is its magnetic dipole moment?
(ii) What is its (approximate) magnetic field at points far from the origin?
(iii) Determine the exact solution for $\vec{B}$ on the positive z axis.

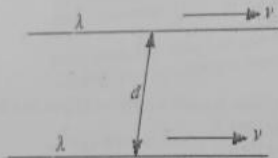
SECTION B

4. A thick slab extending from $z = -a$ to $z = a$ carries a uniform volume current $\vec{J} = J\hat{i}$.
(i) Find the magnetic field both inside and outside the slab.



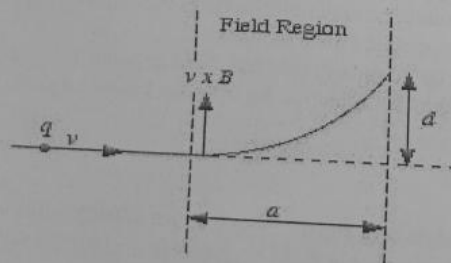
$$\frac{1}{4}\pi\omega R^4$$

- (b) State four configurations that can be handled by Ampere's law
5. (a) With the magnetic field produced by a Steady current. Use Biot- Savart Law to explain.
 (b) Write the two equations for Surface and Volume currents that the Biot- Savart law can be written as.
 (c)



Suppose you have two infinite straight-line charges λ , a distance d apart, moving along at a constant v as above.

- (i) How fast would v have to be in order for the magnetic attraction to balance the electrical repulsion?
6. (a) Explain Ferromagnetic material and give four (4) examples
 (b) A particle of charge q enters the region of uniform magnetic field $\vec{B}$ (pointing into the page). The field deflects the particle a distance d above the original line of flight, as shown in diagram below.
 (i) Is the charge positive or negative?
 (ii) In terms of a , d , B , and q , find the momentum of the particle.



Paper 6 of 6 — PHY 312 2016 2017 ELECTRICITY AND MAGNETISM

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS

FIRST SEMESTER EXAMINATION

Academe Session: 2016/2017

Course Code: PHY 312

Course Title: Electricity and Magnetism I

Course Unit: 3 Unit

Time allowed: 2 hrs 30mins

Instruction: Answer any two (2) questions in section A and any two (2) in section B

SECTION A

QUESTION ONE :

1 (a) A very long, uniform line of charge with positive linear charge density λ lies along the x-axis. An identical line of charge lies along the y-axis.

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(b) Determine the electric field E_y for all points in the x-y plane.

(3) Determine the change in electrostatic potential ΔV between the points $x = 4a$, $y = a$ and $x = a$, $y = 3a$.

(6) Determine ΔV between the points $x = a$, $y = 2$ and $x = 2$, $y = a$.

(c) How much work must be done to move a small negative charge $-q$ from the point $x = 3a$, $y = 3a$ to the point $x = a$, $y = a$?

(b) Which of these magnetic fields can exist? Determine the current density that corresponds to these fields?

(a) $B = e^{-x}$

(b) $B = e^{-x}$

(c) $\mathbf{B} = \sin(x)\mathbf{e}_2$

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7 (a) Define Lorentz Covariance and explain the two distinct but closely related meanings

(b) Maxwell's equations represent one of the most elegant and concise ways to state the fundamental of electricity and magnetism. Discuss

(c) A circular loop of wire, with radius R , lies in the xy plane, centred at the origin, and

carries a current I . What is its magnetic dipole moment?

(d) What is its (approximate) magnetic field at points far from the origin?

(e) Determine the exact solution for $\mathbf{E}$ on the positive z axis

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(a) Show that the magnetic field can not do any work on a particle, i.e. can not change its

: energy.

(b) A spherical shell of conducting material has an inner radius a and an outer radius b . The shell carries no net charge. Inside the shell, located at the centre, is a negative point charge of $-Q$. The shell carries a current J running counter clockwise as viewed from the positive z axis.

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Q. The point charge is suspended from a thin thread and does not touch the shell. The coordinate r measures the distance from the centre of the sphere.

(a) Determine the charges on the inner and outer surfaces of the spherical shell.

(b) Determine the electric field everywhere in space. Sketch $E(r)$ vs. r .

(c) Determine the electrostatic potential $V(r)$ everywhere in space. Sketch $V(r)$ vs. r .

(d) Show that the relation $E = -dv/dr$ is satisfied

SECTION B

QUESTION ONE

(a) Faraday's Law is a fundamental relationship which comes from Maxwell's equations.

Explain

(b) Three very long parallel wires are arranged as shown below. Two wires, which are parallel to the page, pass through the points $(0, 4a)$ and $(0, -4a)$. The third wire carries a current J into the page and is located at the point $x = 2a, y = -a$.

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(c) Determine the magnetic field B at the point $(0, 0)$ due to the other two wires.

(d) Determine the magnetic field B at the point $(0, 0)$ due to the single wire at $x = 2a, y = -a$.

(e) Determine the magnetic field B at the point P due to the two wires located at $x = 0, y = \pm 4a$ at an arbitrary point P on the axis, a distance x from the origin.

(f) Determine the magnetic field B at the point P due to the single wire at $x = 2a, y = -a$.

(g) From (c) and (f), determine the total magnetic field at the point P .

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(h) State and explain the three fundamental quantities of Magnetostatics.

(i) In the diagram below, if F is uniform, show that $\oint \mathbf{F} \cdot d\mathbf{l} = F \oint dl$; where F is the vector from the origin to the point in question. That is check that $\oint \mathbf{x} \cdot d\mathbf{x} = 0$ and $\oint \mathbf{e} \cdot d\mathbf{l} = 0$.

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(j) The three quantities are related and if one of them is known, the other two can be calculated. Draw the table that summarises the relations between them.

(k) State the difference between the Boundary condition for electric field and that of magnetic field.

QUESTION E -

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(b) Two long, straight, parallel current-carrying wires are shown below. The wires are paraliel to the x-axis (the +z-axts poinis out of the page). The current in the wire 'above the origin is flowing in the +z-direction ($I_1 = 3.2\text{A}$, out of the page), while the current in the wire below the origin is owing in the $-z$ -direction ($I_2 = 3.2\text{A}$, imto the page). The wire above the origin passes through $y = +13.3\text{cm}$. the wire below the origin passes through $y = -13.3\text{cm}$, and ihe point P is at $x = +35.0\text{cm}$.

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fo} Want clectic field (siaie all three components) would we need to < 'apply to the electron in part (b) so that the avi force acting on it were zero?

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